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Deep learning short-term load forecasting method based on adaptive mode decomposition

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ABSTRACT

Power load fluctuations are influenced by both natural environmental factors and human activities, exhibiting both periodic and random characteristics simultaneously. The incorporation of temperature-sensitive loads serves to underscore the random character of power load fluctuations. Against this backdrop, short-term power load forecasting has once again come into focus. Extracting the periodic and random fluctuation characteristics of load data is a critical method for improving the accuracy of short-term load forecasting. This paper takes regional load as the research object and proposes a deep learning-based short-term load forecasting method based on adaptive mode decomposition (AMD). First, a feature construction method is proposed that considers regional meteorological differences. Second, a combined time series decomposition method is proposed to extract the stationary and non-stationary components of load fluctuations. Finally, a CNN-LSTM time series neural network model is employed to uncover the mapping relationships between each component and key features. The forecasts of each component are then aggregated to obtain the short-term load forecast. The proposed integrated forecast model is subjected to empirical scrutiny using actual load data from a specific location in Shandong Province. The results demonstrate that the proposed model outperforms traditional models, effectively improving the accuracy of short-term load forecasting.

1. Introduction

Load forecasting is an important guarantee for the safe and economic operation of power systems [1]. Accurate short-term load forecasting has been demonstrated to optimize grid dispatching, facilitate the intelligent and digital construction of power systems, and play a pivotal role in energy management [2,3]. Due to the multiple influences of meteorological conditions and social activities on power load, its periodicity and randomness are coupled, which increases the difficulty of power load forecasting. Therefore, there is an urgent need to propose advanced forecasting methods tailored to the complex fluctuations of power load to improve the accuracy of power load forecasting. With the rapid development of artificial intelligence technology, deep learning methods have been widely applied due to their powerful nonlinear fitting and generalization capabilities [4]. The utilization of a solitary forecasting model will limit deep learning's ability to extract data volatility and nonlinearity. The utilization of a diverse array of deep learning models and the construction of deep learning combinatorial variant methods can give full play to the advantages of various models and more comprehensively mine data sample features [5]. The non-stationarity of data samples is more significant due to the deep coupling of power load periodicity and randomness. The load data is preprocessed by a data decomposition method to reduce the complexity of the load sequence. Then, each subsequence is forecasted independently, a process that has the potential to further enhance the generalization capability of deep learning and improve the forecasting accuracy [6].

Experts and scholars at home and abroad have proposed a variety of short-term load forecasting models, mainly divided into traditional statistical methods such as time series methods [7,8], autoregressive

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models [9,10] and machine learning models such as support vector machine (SVM) [11], extreme learning machine [12], random forests [13] and mechanical learning models such as extreme gradient boosting trees, but the above statistical methods have limited ability to deal with complex non-linear problems and can hardly capture the characteristic relationships between load data and relevant influencing factors comprehensively, which makes it more difficult to improve the forecasting accuracy.

With the rapid development of artificial intelligence technology, deep learning methods have been widely used in power system fault diagnosis [14], smart grid [15], distribution network reconstruction [16], state estimation [17], load forecasting [18], new energy prediction [19], and other fields because of their powerful nonlinear fitting ability and generalization capability. In recent years, especially in the field of load forecasting has received much attention from scholars at home and abroad. The literature [20] and the literature [21] used back propagation neural network (BPNN) and recurrent neural network (RNN) to forecast the load variation of the power system, respectively, which effectively improved the accuracy of load forecasting. However, the modeling process of BPNN and RNN ignores the time-series correlation between historical data, and the accuracy of load forecasting is limited. The long short-term memory network (LSTM) is an improved model based on RNN, which overcomes the gradient explosion problem inherent in RNN and can effectively exploit the time-series characteristics of data to improve the forecasting accuracy, and has been quite successful in the field of load forecasting. The literature [22] firstly deals with the load data and then constructs an LSTM model to forecast the load, and obtains better forecasting results. In [23], a convolutional neural network (CNN)-LSTM-based load forecasting model was proposed, and the results showed that the CNN-LSTM model had superior forecasting performance compared with the LSTM model.

In the actual operation of power systems, load is affected by many complex factors, such as temperature, wind speed, and date type. At the same time, with the widespread connection of a large number of user-side distributed photovoltaic systems, the power load has become extremely uncertain. Using a single prediction model alone cannot effectively eliminate the impact of the original load's non-stationarity on the forecasting results, making it difficult to obtain accurate forecasting [24]; the combination of forecasting methods can give full play to the advantages of different models and methods, and more comprehensively mine the characteristics of the dataset, thus obtaining more accurate forecasting results. To cope with the influence of external factors, the load data are preprocessed through data decomposition methods to reduce the complexity of the load sequence, and then the sub-sequences are modelled and predicted separately. The final forecasting results are obtained by superimposing the predicted values of the sub-sequences, which can effectively improve the accuracy of the load forecasting [25]. Among them, literature [26] used wavelet decomposition to process the load data for forecasting, but the wavelet decomposition [27] has the problem of difficult selection of basis functions and thresholds, which makes it difficult to improve the load forecasting accuracy effectively. For this reason, literature [28] and [29] used empirical mode decomposition (EMD) and ensemble empirical mode decomposition (EEMD), respectively, to process load data and perform forecasting, which improved the load forecasting accuracy to some extent. Forecasting accuracy is achieved to a certain extent; however, the decomposition of the EMD and EEMD methods is plagued by spectral aliasing, resulting in large forecasting errors due to the high-frequency, strong non-stationary components generated during the decomposition.

To this end, this paper proposes a deep learning-based short-term load forecasting method employing adaptive mode decomposition, tailored to the fluctuating characteristics of loads during non-holiday periods. First, a feature construction method considering regional meteorological differences is established using numerical weather prediction, enabling in-depth analysis of key meteorological factors influencing load fluctuations. Then, the power load data samples are decomposed using improved complete ensemble empirical mode decomposition with adaptive noise. Sample entropy is employed to calculate the temporal complexity of each component. The components are aggregated and reconstructed based on complexity similarity. This approach enables the representation of the periodicity and randomness of load fluctuations through stationary and non-stationary components. Consequently, it facilitates the computational process of the model while establishing conducive conditions for enhancing the forecasting performance of the model. Finally, a deep feature mining CNN-LSTM time series network model has been developed, integrating the strengths of CNN and LSTM. The CNN fully extracts the high-dimensional meteorological characteristics that significantly impact load component fluctuations, while the LSTM accentuates the role of crucial time series data, thereby enhancing the forecasting accuracy of power load. This method effectively characterizes the fluctuating characteristics of load fluctuations, deeply extracts the complex nonlinear high-dimensional features influencing load fluctuation patterns, and addresses the prominent issue of the high difficulty in load forecasting caused by the combined effects of strong volatility, periodicity, and randomness in power loads.

2. Load sequence decomposition method based on AMD

This section introduces an adaptive mode decomposition (AMD) method to extract the stationarity and randomness of load sequences. The method decomposes load sequences into distinct modal components. This process simultaneously reduces sequence complexity. Fig. 1 illustrates the AMD-based decomposition process.

AMD integrates Improved Complete Ensemble Empirical Mode Decomposition with Adaptive Noise (ICEEMDAN). This method applies ICEEMDAN to decompose load data, separating effective signals while removing noise. This process identifies load sequence stationarity and non-stationarity, reducing data complexity. Subsequently, the time sequence complexity of several components is calculated based on the sample entropy method. Components with similar complexity are clustered to reduce component count and computational burden.

2.1. Improved complete ensemble empirical mode decomposition with adaptive noise

ICEEMDAN is a novel data decomposition technology that represents an enhancement to the conventional EMD. It employs adaptive noise and full integration strategies to enhance decomposition stability and accuracy [30]. The ICEEMDAN method involves several distinct steps. ICEEMDAN first decomposes the original signal into multiple Intrinsic Mode Functions (IMFs) via EMD. It then applies adaptive noise removal to each IMF before fully integrating the denoised components. Compared to conventional EMD, this method's adaptive noise reduction suppresses mode mixing while complete integration preserves critical signal features, significantly improving decomposition accuracy.

EMD begins by identifying local extrema in the original data. These points construct upper and lower envelopes via smoothing curves, with their mean forming the first IMF. Subtracting this IMF from the original signal yields a residual, which becomes the new input for iterative decomposition until meeting the Cauchy convergence criterion. This process generates a complete set of IMFs and residuals capturing the data's local characteristics, governed by the relationship:

$$\begin{cases} r_1(t) = x(t) - IMF_1(t) \\ r_k(t) = r_{k-1}(t) - IMF_k(t) \end{cases} \tag{1}$$

where, $x(t)$ is the original time series; $IMF_1(t)$ is the first component sequence; $r_1(t)$ is the first residual error sequence; $r_k(t)$ is the k residual error sequence; and $IMF_k(t)$ is the k component sequence.

The uneven distribution of extreme points in time series samples distorts the upper and lower envelopes during EMD. This distortion causes significant variations in the time scales of the resulting IMF

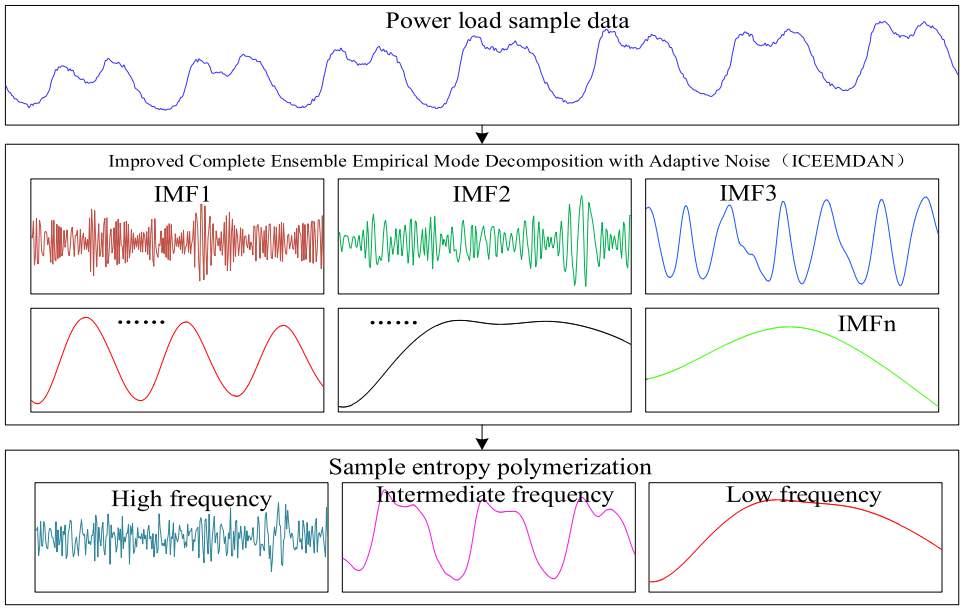


Fig 1. Power load data decomposition process based on AMD.

components and makes EMD susceptible to mode mixing. Colominas et al. proposed that ICEEMDAN leverages the property that the mean value of white noise is 0 [31]. By introducing white noise into the decomposition process, ICEEMDAN effectively eliminates the original signal noise, thereby obtaining more accurate upper and lower boundary lines. The overall idea of ICEEMDAN has been adjusted, and the residual error is obtained as a priority. Obtain the IMF component sequence. To this end, the operator of $M(\cdot)$ producing the local mean and the operator of $E_k(\cdot)$ producing the k IMF by EMD are introduced. ICEEMDAN's sequence decomposition process is shown in Fig. 2.

The specific decomposition process is as follows:

1) Add $E_1(\omega^i(t))$ to the original $x(t)$, then there is:

$$x^i(t) = x(t) + \varepsilon_0 E_1(\omega^i(t)) \tag{2}$$

where i=1,2,3,...,n; $\omega^i(t)$ is the i-th white noise to be added; ε_0 is the signal-to-noise ratio; n is the amount of added white noise; $x^i(t)$ is the i-th signal to be constructed.

2) Calculate the local mean $M(x^i(t))$ of $x^i(t)$ through EMD, then the first residual is shown in equation $r_1(t)$ and the first IMF component $IMF_1(t)$ is shown in Eq. (3).

$$\begin{cases} r_k(t) = \frac{1}{n}\sum_{i=1}^{n} M(x^i(t)) \\ IMF_1(t) = x(t) - r_1(t) \end{cases} \tag{3}$$

3)The kth $IMF_k(t)$ component can be obtained by recursive looping as in Eq. (4)

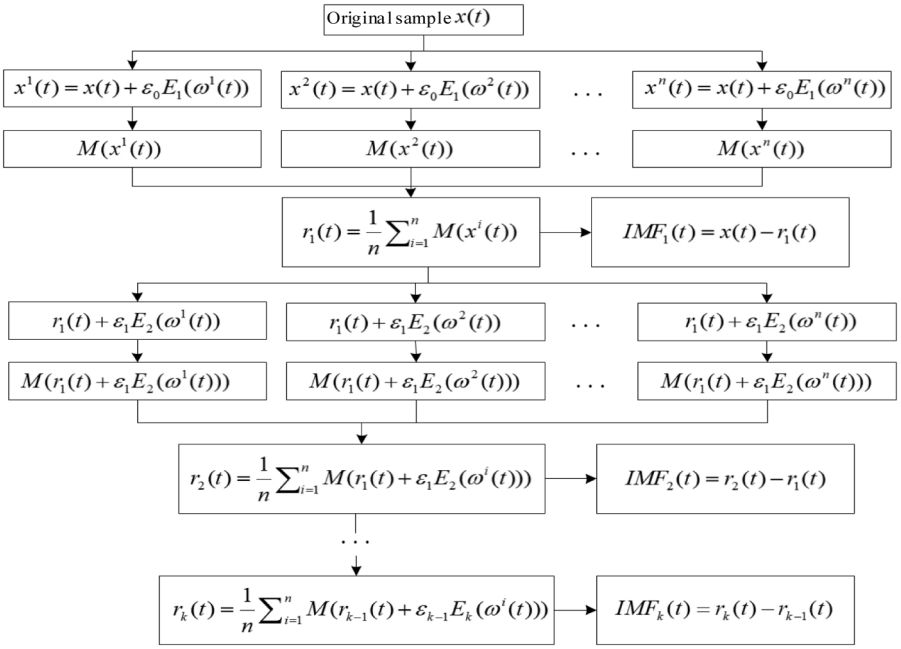


Fig. 2. Time series decomposition process based on ICEEMDAN.

$$\begin{cases} r_k(t) = \frac{1}{n}\sum_{i=1}^{n} M\left(r_{k-1}(t) + \varepsilon_{k-1}E_k\left(\omega^i(t)\right)\right) \\ IMF_k(t) = r_{k-1}(t) - r_k(t) \end{cases} \tag{4}$$

where,$k \geq 2$.

2.2. Sample entropy

While ICEEMDAN significantly improves forecasting model accuracy when processing original load sequences, the increased number of resulting IMF components raises computational demands. Repeatedly modeling highly similar components creates an unnecessary computational burden. Aggregating components of comparable complexity can effectively reduce model construction complexity. Therefore, this paper employs the sample entropy method to quantify the time complexity of each IMF component, enabling the aggregation and reconstruction of components with similar complexity. Sample entropy measures time series complexity by calculating the probability of new pattern generation within the signal [32]. It offers two key advantages: independence from data length and superior consistency. Crucially, a lower sample entropy indicates higher time series self-similarity, while a higher value corresponds directly to greater complexity [33]. The calculation process is as follows:

In general, for a time series $X(i) = [x(1), x(2), \cdots, x(n)]$ consisting of n data points, sample entropy is calculated as follows:

(1) Form a set of vector sequences with dimension m by serial number$X_m(1), \cdots, X_m(n-m-1)$, where $X_m(i) = \{x(i), x(i+1), \cdots, x(i+m-1)\}$,$1 \leq i \leq N-m+1$. These vectors represent the values of m consecutive x's starting at point i.

(2) Define the distance $d[X_m(i), X_m(j)]$ between vectors $X_m(i)$ and $X_m(j)$ as the absolute value of the largest difference in the corresponding elements of the two. To wit:

$$d[X_m(i), X_m(j)] = \max_{k=0,\cdots,m-1}(|x(i+k) - x(j+k)|) \tag{5}$$

(3) For a given $X_m(i)$, count the number of $j(1 \leq j \leq N-m, j \neq i)$ whose distance between $X_m(i)$ and $X_m(j)$ is less than or equal to r, and write it as B_i. For $1 \leq j \leq N-m$, define:

$$B_i^m(r) = \frac{1}{N-m-1} \tag{6}$$

(4) Define $B^{(m)}(r)$ as:

$$B^{(m)}(r) = \frac{1}{N-m}\sum_{i=1}^{N-m} B_i^{(m)}(r) \tag{7}$$

(5) Add dimension to $m+1$, count the number of distances between $X_{m+1}(i)$ and $X_{m+1}(j)(1 \leq j \leq N-m, j \neq i)$ that are less than or equal to r, denoted as A_i. $A_i^m(r)$ is defined as:

$$A_i^m(r) = \frac{1}{N-m-1}A_i \tag{8}$$

(6) Define $A^m(r)$ as:

$$A^m(r) = \frac{1}{N-m}\sum_{i=10}^{N-m} A_i^m(r) \tag{9}$$

Thus, $B^m(r)$ is the probability that two sequences match m points under a similar tolerance r, and $A^m(r)$ is the probability that two sequences match $m+1$ points. Sample entropy is defined as:

$$SampEn(m, r) = \lim_{N\to\infty}\left\{-\ln\left[\frac{A^m(r)}{B^m(r)}\right]\right\} \tag{10}$$

When N is finite, it can be estimated using the following formula:

$$SampEn(m, r, N) = -\ln\left[\frac{A^m(r)}{B^m(r)}\right] \tag{11}$$

3. CNN-LSTM deep learning forecasting model

3.1. Convolutional neural networks

Convolutional neural network models effectively extract key features from input data. This capability stems from local perception and weight sharing, mapping high-dimensional inputs to low-dimensional spaces [34]. The structure of the CNN model is shown in Fig. 3.

The convolutional layer employs shared convolution kernels to perform convolution operations. Each kernel convolves over feature maps from the preceding layer, generating new feature maps. Thus, multiple distinct kernels extract diverse feature maps within the convolutional layer. The convolutional layer utilizes shared weights for efficient feature extraction. By controlling the size and number of convolution kernels, the complexity of feature extraction of high-dimensional data can be significantly reduced. The convolutional layer is calculated as follows: (12)

$$C_i^l = f\left(I^{l-1} \otimes K_i^l + B_i^l\right) \tag{12}$$

where, I^{l-1} is the feature set of layer $l-1$; K_i^l is the i convolution kernel of layer l; B_i^l the bias between the $l-1$th layer input and the ith feature set of the lth layer; $f(\cdot)$ is the activation function of this layer; and this paper chooses the Tanh function as the activation function.

Convolutional Layer divides each feature set of the upper layer into several different feature subsets of the same dimension. The pooling layer uses the downsampling method to further reduce the data and extract the features of each feature subset. A new set of low-dimensional features equal in number to the convolutional layer can be formed. The mathematical expression of the pooling layer is (13).

$$P_i^{l+1} = f\left(\alpha_i^{l+1} \cdot down\left(C_i^l\right) + B_i^{l+1}\right) \tag{13}$$

where, α_i^{l+1} is the weight coefficient of the ith feature set of the $l+1$th layer; C_i^l denotes the ith feature subset of the lth convolutional layer; B_i^{l+1} is the bias between the $l+1$th layer and the ith feature set of the lth layer; $down(\cdot)$ is the downsampling function, generally maximum downsampling and mean downsampling; P_i^{l+1} is the ith feature subset of the $l+1$th pooling layer.

The fully connected layer is consistent with the traditional neural network, which connects each neuron with all the neurons in the previous layer and integrates all the extracted features. The mathematical expression is (14).

$$f_i = f\left(\sum_{i\in l+1} W_i \cdot P_i^{l+1} + b_i\right) \tag{14}$$

where f_i is the ith neuron; W_i is the ith weight parameter, which determines the dimensionality based on each feature subset of the pooling layer, which is generally one-dimensional or multidimensional; and b_i is the bias.

3.2. Long short-term memory neural networks

1)Recurrent Neural Network

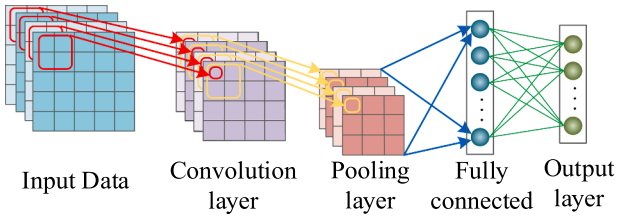


Fig. 3. Diagram of CNN.

Due to the continuity of atmospheric motion and complex social activities, the load must have a certain relationship in time, so the traditional neural network will weaken the effect of load forecasting. Paul Werbos et al. proposed the RNN, which differs from feedforward architectures through specialized cyclic units that incorporate historical information into current time-step processing [35].

With Tanh as neuron A's activation function, this recurrent unit forms a cyclic structure.

$$c_t = \tanh(W \cdot c_{t-1} + V \cdot x_t] + b_r) \tag{15}$$

$$h_t = V \cdot c_t \tag{16}$$

wherec_t is the memory information of the current time t; c_{t-1} is the memory information of the previous time t-1,b_r is the bias, x_t is the input feature at time t; h_t is the output value of the cyclic unit at time t; and V and W are the relevant parameters.

Learning long-term information involves multiple iterations of the above process, and if the historical data is distant from the current data, the number of iterations increases. When training neural networks using the gradient descent method, it can cause the continuous multiplication of partial derivatives, which may lead to the problem of "gradient vanishing". Typical recurrent neural networks are not suitable for learning long-term memory information [36].

2)Long Short-Term Memory Neural Networks

Long short-term memory neural networks are developed from recurrent neural networks, which offer distinct advantages in processing long-term information and can avoid the problem of "gradient vanishing". Its main structure is a cyclic unit that contains several special multiplication units, called "gates", which control the timing state of information flow in the network [37,38]. The input gate controls the activation process of the input, which contains the current input data and the output value of the preceding time information. The output gate controls the activation process of the output. The forget gate controls the degree of forgetting of the memory unit's information at the last moment. The three gates collaborate to eliminate invalid information and retain beneficial information.

First, the forget gate reads the output data from the previous time and the input data from the current time. It then obtains an attenuation coefficient using a sigmoid function as the activation function. This coefficient is then used to calculate the memory unit from the previous time. The process is described as follows:

$$f_t = \sigma\left(w_{f1} \cdot h_{t-1} + w_{f2} \cdot x_t + b_f\right) \tag{17}$$

where, σ is the sigmoid function; w_{f1} and w_{f2} are the weights; b_f is the bias; f_t is the attenuation coefficient, and $f_t \in [0,1]$; x_t is the current input value; and h_{t-1} is the last moment output value.

Similarly, the input gate process is described as:

$$i_t = \sigma(w_{i1} \cdot h_{t-1} + w_{i2} \cdot x_t + b_i) \tag{18}$$

$$\widetilde{C}_t = \tanh(w_{c1} \cdot h_{t-1} + w_{c2} \cdot x'_t + b_c) \tag{19}$$

where, w_{i1}, w_{i2}, w_{c1} and w_{c2} are the weights;b_i and b_c are bias; $\widetilde{C}_t$ is the temporary memory unit.

The current memory unit can be obtained from the memory unit of the previous moment and the temporary memory unit:

$$C_t = f_t \cdot C_{t-1} + i_t \cdot \widetilde{C}_t \tag{20}$$

Finally, information output is realized by the output gate:

$$o_t = \sigma(w_{o1} \cdot h_{t-1} + w_{o2} \cdot x_t + b_0) \tag{21}$$

$$C_t = f_t \cdot C_{t-1} + i_t \cdot \widetilde{C}_t \tag{22}$$

3.3. Timing model of CNN-LSTM depth feature mining

In this section, high-dimensional features are mined by using the CNN's strong capability in high-dimensional data processing, while temporal features are captured through the LSTM network.The CNN-LSTM model consists of five layers, as shown in Fig. 4. The specific structure is as follows:

1) Input layer: The gridded numerical weather forecasting feature table is input into the convolutional neural network as the feature variable.
2) CNN layer: Consisting of a four-dimensional convolution layer, a two-dimensional pooling layer, and a fully connected layer, it is used to extract key spatial and meteorological features from the input data. The Tanh function is selected as the activation function for the convolution layer, and the maximum pooling mode is used for feature reduction. The meteorological feature vector is obtained after convolution and pooling operations.
3) Fully connected layer: The historical load sequence and meteorological feature vector are combined as the input of the fully connected layer, enhancing the influence of load component autocorrelation and preventing information loss when convolving with the high-dimensional meteorological input. The Tanh activation function is also adopted at this layer.
4) LSTM layer: This layer learns the internal relationships between the CNN-extracted feature vectors, load components, and temporal variation patterns, and predicts future load values. To prevent overfitting and enhance the model's generalization ability, Dropout layers are incorporated within the CNN-LSTM framework to randomly deactivate neurons during training, reducing co-adaptation among units. In addition, early stopping is employed to terminate training when the validation loss no longer decreases after several epochs, avoiding over-training. Furthermore, a learning rate decay strategy is applied to ensure smooth and stable convergence during the optimization process.
5) Output layer: The Tanh function is selected as the activation function. The final forecasting result of each load component is calculated based on the output of the LSTM layer.

4. Deep learning short-term load forecasting model based on adaptive mode decomposition

To improve the accuracy of load forecasting, the adaptive mode decomposition deep learning short-term load forecasting method proposed in this paper is mainly divided into four steps:

1) Raw load data processing. Using the AMD method to decompose the original load data, N relatively stable component sets {SIMF1, SIMF2, ..., SIMFn}.

2) Data pre-processing. Each component was normalized to the meteorological data by:

$$f' = \frac{f - f_{\min}}{f_{\max} - f_{\min}} \tag{23}$$

where f andf' are the data before and after normalization, respectively; $f_{\min}$ and $f_{\max}$ are the minimum and maximum values of the data set, respectively.

3) The adaptive mode decomposition deep learning forecasting model is constructed. A forecasting model is established for each decomposed component. The historical data and meteorological characteristic data of each component are used as input to train the model. The model parameters are iteratively optimized using a loss function, and the forecasting results for each component are output.

4) Analysis of forecasting results. The load forecasting results are obtained by superimposing the forecasting results of each component, and the forecasting performance of the proposed model is analyzed by

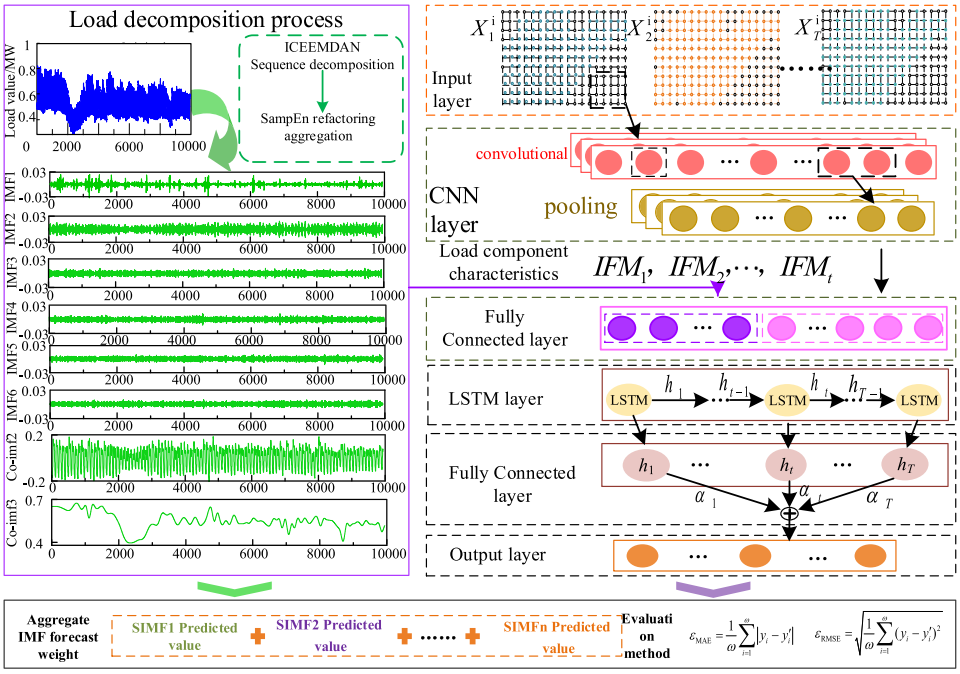


Fig. 4. Timing model of CNN-LSTM depth feature mining: Case analysis.

comparing the error evaluation index with the forecasting results of the comparison model.

4.1. Data selection and model testing environment

In this paper, we use the Jupyter integrated development environment in Anaconda software and the TensorFlow deep learning framework to build a short-term load forecasting model 24 hours ahead of time. The numerical weather forecast data and measured load data of a place in Shandong Province, every 15 minutes from January 1, 2019, to December 31, 2021, were selected, and all the data were divided into three sets. The relevant data from January 1, 2019, to December 31, 2019, was used for model training. January 1, 2020, to December 31, 2020, for model testing and January 1, 2021, to December 31, 2021, for model forecasting. The training and forecasting process of this model is based on the following environment: Windows 11, inteli7-10875H@2.30GHz; Nvidia RTX2060; 32GB of running memory.

To improve the reliability of model training and parameter tuning, cross-validation was applied to the dataset from 2019-2020. This procedure ensured that all available data were effectively utilized for both training and validation, enhancing the robustness of the model and preventing overfitting.

4.2. Evaluation indicators

In this paper, the mean absolute error (MAE), root mean square error (RMSE), and coefficient of determination (R^2) are selected as the primary evaluation indicators to characterize the model forecasting accuracy. MAE, RMSE, and R^2 are calculated by the following equations:

$$\varepsilon_{\mathrm{MAE}} = \frac{1}{\mathrm{T}}\sum_{i=1}^{\mathrm{T}}|y_i - y'_i| \tag{24}$$

$$\varepsilon_{\mathrm{RMSE}} = \sqrt{\frac{1}{\mathrm{T}}\sum_{i=1}^{\mathrm{T}}(y_i - y'_i)^2} \tag{25}$$

$$\varepsilon_{R^2} = 1 - \left[\sum_{i=1}^{\mathrm{T}}(y_i - y'_i)^2 \Big/ \sum_{i=1}^{\mathrm{T}}(y_i - \bar{y})^2\right] \tag{26}$$

Where T is the predicted time duration; y_i and y'_i are the actual and forecast values of the normalized load at time i, respectively; and $\bar{y}$ is the average value of the normalized load.

4.3. Validity verification of load decomposition

(1) Adaptive mode decomposition

In this paper, ICEEMDAN was used to decompose the load sequence, and 12 IMF components were obtained. Fig. 5 shows the load decomposition from December 1 to December 3, 2019. As illustrated in the figure, the ICEEMDAN model effectively decomposed the stationary and non-stationary sequences contained in the original load sequence. The load sequence components of partial decomposition exhibit a high degree of similarity. The modeling of each component will undoubtedly yield a phenomenon that is the focus of a significant amount of modeling work aimed at enhancing accuracy.

Therefore, it is necessary to reasonably merge the components with higher similarity to reduce the number of components and the modeling workload. To this end, the sample entropy proposed in this paper is used to calculate the timing complexity of each component. Components with similar timing complexity are superimposed and aggregated to reduce the number of components that are ultimately used for modeling. The results of the calculation of the complexity similarity of each component are shown in Fig. 6.

As demonstrated in Fig. 6, the component sequence was artificially aggregated based on the number of components present in each superposition component. Furthermore, the IMF component with high entropy of the first three samples was superimposed into a post-processing component, SIMF1. Components IMF4, IMF5, and IMF6 were superimposed to obtain SIMF2, and the remaining component, IMF, was merged into SIMF3 at last. The selection of component superposition is shown in Table 1, and Fig. 7 shows the load component after superposition. It can be seen that the stationarity of the SIMF1 component is poor, while the SIMF2 and SIMF3 components are relatively stable.

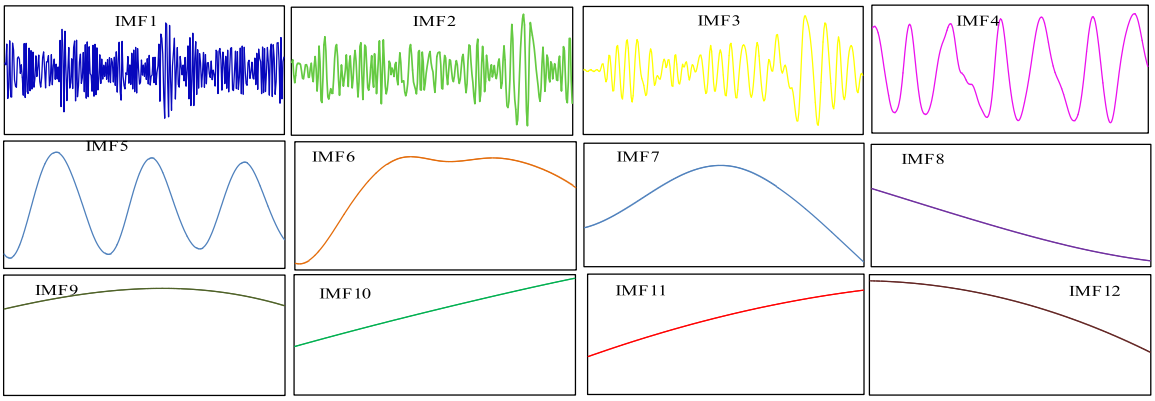


Fig. 5. Load breakdown of a place in Shandong Province from December 1 to December 3, 2019.

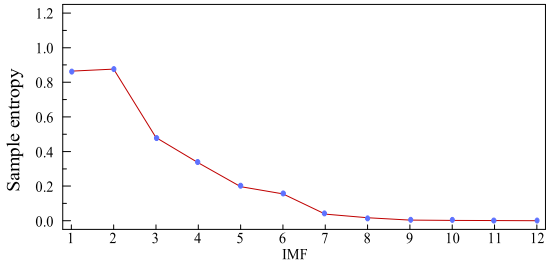


Fig. 6. Sample entropy of each component sequence.

Table 1
IMF weight stack results.

Serial number	1	2	3
Original IMF component serial number	1, 2, 3	4, 5, 6	7, 8, 9, 10, 11, 12

(2) Comparison of forecasting results of different components

To verify the validity of forecasting results for each type of component, the proposed CNN-LSTM model was used to predict SIMF1, SIMF2, and SIMF3 components after superposition reconstruction. Table 2 presents the daily average forecasting accuracy of SIMF1, SIMF2, and SIMF3 components on the selected day. The monthly mean MAE error of the SIMF1 component is 4.19 % and the RMSE error is 5.05 %. Similarly, the monthly mean MAE error of the SIMF2 component is 1.07 % and the RMSE error is 1.36 %. The monthly mean MAE error of the SIMF3 component is 3.52 %, and the RMSE error is 4.35 %, indicating that the forecasting effect of the SIMF1 component is relatively poor. While IMF3 demonstrates enhanced stability to SIMF2, it exhibits notable deviations from the input variable's shape, resulting in a forecasting effect that is marginally inferior to that of SIMF2.

(3) Analysis of short-term load forecasting results

To validate the superiority of the proposed method (M5), four benchmark models were implemented for comparison: support vector machine (SVM, M1), BPNN (M2), extreme gradient boosting (M3), and CNN-LSTM without decomposition (M4). In this paper, the load forecasting accuracy of the 12 months in 2021 is compared with that of the four models. The results of the monthly average accuracy are shown in Tables 3 and 4.

As shown in Tables 3 and 4, the temporal comparison indicates that the forecast accuracy of the first quarter(January to March) was moderate, lower than that of the fourth quarter, at 1.82 %, 1.06 %, and 1.25 %, respectively. The forecast accuracy for the second quarter(April to June) is relatively high, at 1.24 %, 1.35 %, and 1.36 % respectively. The forecast accuracy of the third quarter (July to September) is relatively low, at 2.56 %, 2.77 % and 2.87 % respectively. The forecast accuracy of the fourth quarter (October to December) is slightly lower than that of the second quarter, at 1.84 %, 1.97 % and 1.72 % respectively. The main reason is that during the second quarter, weather conditions are most

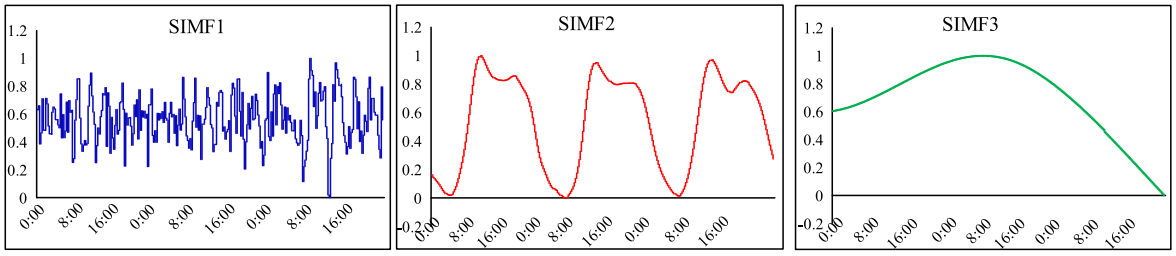


Fig. 7. The SIMF components obtained from ICEEMDAN decomposition are obtained by superposition.(2).

Table 2
The MAE and RMSE values of each model on a certain day from January to December.

Error index	SIFM	Jan.	Feb.	Mar.	Apr.	May.	Jun.	Jul.	Aug.	Sept.	Oct.	Nov.	Dec.	mean value
MAE (%)	SIFM1	2.16	2.17	1.92	4.57	3.72	7.48	1.32	1.62	6.35	6.36	4.36	8.20	4.19
	SIMF2	0.89	0.77	0.56	1.47	1.48	0.84	0.42	0.99	0.71	1.86	1.32	1.58	1.07
	SIMF3	1.64	4.76	2.49	3.89	2.20	1.26	14.67	2.09	1.61	2.33	3.60	1.70	3.52
RMSE (%)	SIFM1	2.72	2.61	2.34	5.68	4.86	8.68	1.74	1.98	7.14	7.57	5.22	10.03	5.05
	SIMF2	1.16	1.06	0.76	1.72	1.80	1.05	0.55	1.27	0.87	2.37	1.62	2.16	1.36
	SIMF3	2.08	5.72	3.06	4.71	2.61	1.59	17.95	2.75	1.93	2.92	4.50	2.36	4.35

Table 3
Comparison of the forecasting accuracy of MAE values of each model from January to December.

Models	Jan.	Feb.	Mar.	Apr.	May.	Jun.	Jul.	Aug.	Sept.	Oct.	Nov.	Dec.
M1	2.74	3.29	3.30	2.98	3.07	3.21	4.33	4.75	5.07	3.78	3.01	3.54
M2	2.47	2.62	2.96	2.76	2.60	3.15	3.38	3.54	3.63	2.01	2.52	2.10
M3	2.25	1.94	2.20	1.76	1.72	2.05	3.36	3.44	3.49	2.22	2.92	2.81
M4	2.60	1.89	1.83	1.49	1.46	1.52	3.12	3.85	3.00	1.70	2.04	2.04
M5	1.99	1.58	1.69	1.06	1.15	1.18	2.33	2.13	2.29	1.36	1.50	1.14

Table 4
Comparison of the forecasting accuracy of RMSE values of each model from January to December.

Models	Jan.	Feb.	Mar.	Apr.	May.	Jun.	Jul.	Aug.	Sept.	Oct.	Nov.	Dec.
M1	4.38	3.66	3.89	3.17	3.46	3.90	4.76	5.29	5.48	4.51	3.62	3.90
M2	3.59	2.80	2.71	2.91	2.86	3.84	3.85	3.99	4.22	2.96	3.15	2.61
M3	3.11	2.29	2.41	2.36	2.28	2.46	3.65	3.71	4.01	2.39	3.34	3.08
M4	2.29	1.95	2.05	1.81	1.76	1.97	3.63	4.15	3.56	1.79	2.41	2.67
M5	1.82	1.06	1.25	1.24	1.35	1.36	2.56	2.77	2.87	1.84	1.97	1.72

stable, the environment is suitable, and the impact of human production activities is minimal, resulting in relatively stable power load fluctuations that are easier to predict. In the third quarter, weather conditions changed significantly, temperatures were high, and power loads fluctuated sharply due to the cooling load, making forecasts challenging. During the winter months of the first and second quarters, the load is influenced by a specific heating load. However, due to the uniform heating requirements across China, the cooling load exhibits less variability, thereby enhancing the forecast accuracy compared to that of the summer months in the third quarter.

Analyzing the monthly average forecasts, derived from the MAE values, revealed a decline in the average forecasted MAE of model M5 by 1.97 %, 1.19 %, 0.89 %, and 0.59 %, respectively, in comparison to models M1, M2, M3, and M4. The analysis of the RMSE values indicates a decline in the average predicted RMSE values of the M5 model by 2.38 %, 1.47 %, 1.1 %, and 0.68 %, respectively, in comparison to the models M1, M2, M3, and M4. To provide a visual representation of the monthly average forecasting accuracy of the M5 model, the RMSE and MAE metrics are represented as in Fig.8.

To further illustrate the forecasting effect of the M5 model proposed in this paper, the forecasting results of one randomly selected day from each month of 2021 are analyzed. As shown in Fig. 9, the forecasting of the randomly selected days from January to December, respectively, demonstrates the efficacy of the model. Table 5 presents the average daily load forecast accuracy for the specified days.

5. Conclusion

This paper proposes a short-term load forecasting method based on the AMD-CNN-LSTM model, with the following conclusions drawn from the analysis of arithmetic examples:

(1) The proposed AMD method effectively reduces the irrelevant noise in high-frequency components, providing smoother training data for forecasting models and improving accuracy. Compared to the single decomposition method, the MAE and RMSE are reduced by 27.72 MW and 34.90 MW, respectively, and the R^2 is improved from 0.33 to 0.95.
(2) The CNN-LSTM model can give full play to the capability of CNN feature extraction, and combined with the time series memory and long short-term dependence processing capability of LSTM model, the correlation between meteorological data and time series output can be deeply mined, and the accuracy of the time series forecast can be improved.
(3) Simulation results show that the proposed AMD-CNN-LSTM model exhibits superior average MAE, RMSE, and R^2 values in comparison to the other four models. This suggests that the AMD-CNN-LSTM model has the potential to provide more accurate forecasts and offer promising applications in the domain of short-term load forecasting.

CRediT authorship contribution statement

Yixiao Yu: Writing – original draft. **Fei Wang:** Writing – original draft. **Ming Yang:** Resources. **Yumin Zhang:** Software. **Zhenyu Wang:** Methodology.

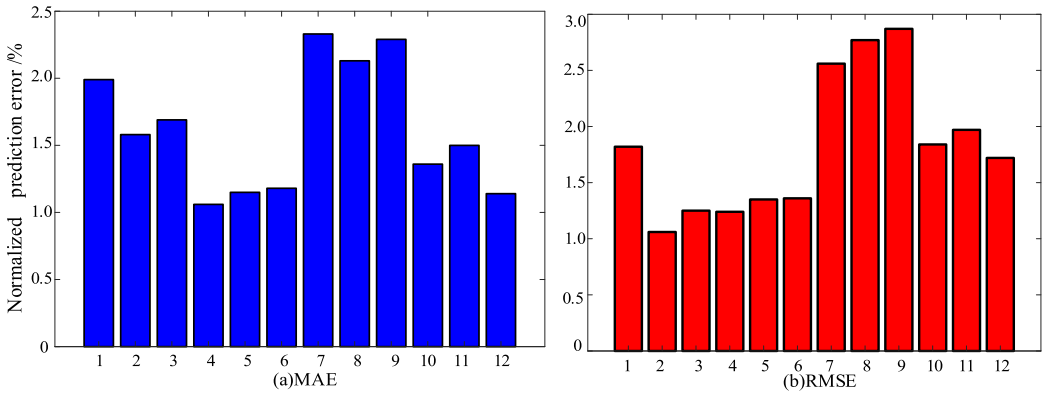


Fig. 8. MAE (a) and RMSE (b) monthly mean predictions of model M5 proposed in this paper.

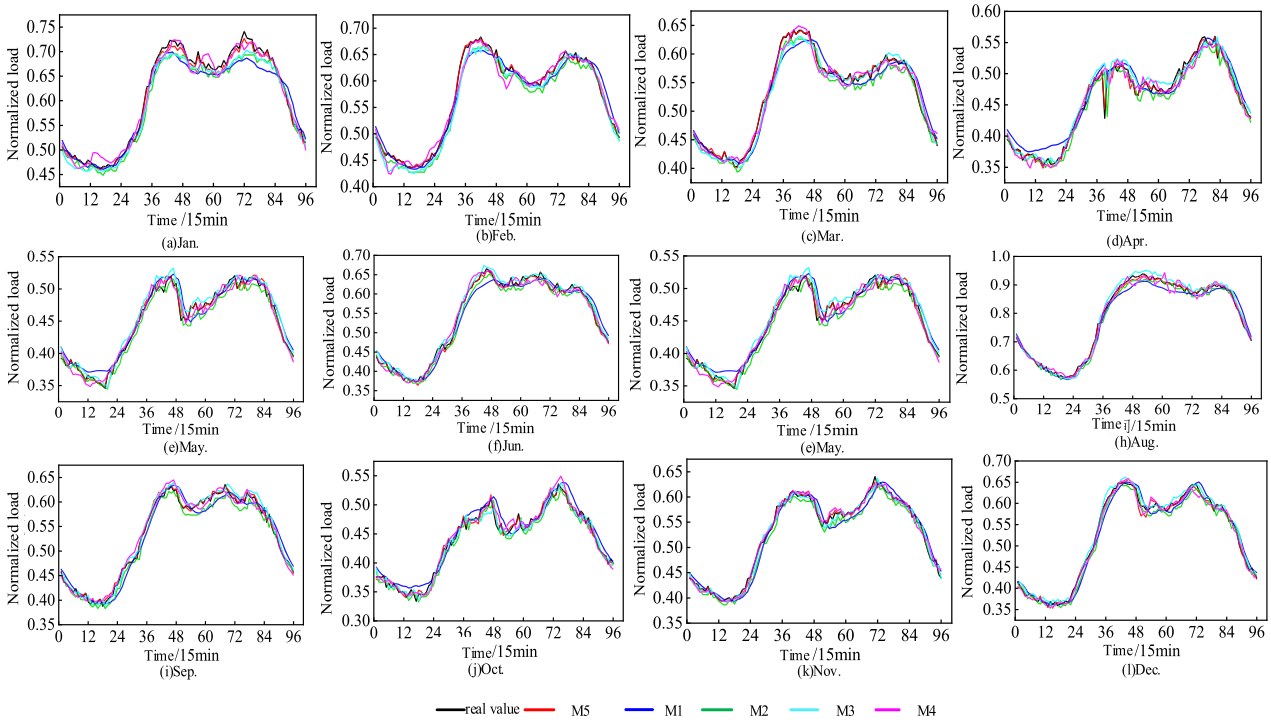


Fig. 9. Daily load forecast from January to December.

Table 5
Comparison of forecasting accuracy of each model from January to December.

Error index	Models	Jan.	Feb.	Mar.	Apr.	May.	Jun.	Jul.	Aug.	Sept.	Oct.	Nov.	Dec.
MAE (%)	M1	2.33	1.65	1.95	2.15	1.76	2.14	2.01	1.70	1.93	2.01	1.70	1.93
	M2	2.27	1.79	1.78	1.90	1.70	1.69	1.67	1.33	1.78	1.67	1.33	1.78
	M3	1.95	1.59	1.48	1.73	1.70	1.34	1.12	1.18	1.30	1.12	1.18	1.30
	M4	1.53	1.38	1.36	1.47	1.29	1.27	0.96	1.11	1.20	0.96	1.11	1.20
	M5	0.74	0.32	0.35	0.73	0.62	0.67	0.33	0.44	0.45	0.33	0.44	0.45
RMSE (%)	M1	2.97	2.10	2.47	2.89	2.38	2.62	2.51	2.06	2.31	2.51	2.06	2.31
	M2	2.73	2.16	2.11	2.79	2.10	2.11	2.01	1.61	2.16	2.01	1.61	2.16
	M3	2.38	2.03	1.84	2.52	2.17	1.64	1.40	1.45	1.64	1.40	1.45	1.64
	M4	1.88	1.88	1.68	2.28	1.66	1.66	1.29	1.44	1.50	1.29	1.44	1.50
	M5	0.86	0.41	0.42	0.92	0.80	0.84	0.40	0.56	0.56	0.40	0.56	0.56
R^2	M1	0.948	0.968	0.954	0.931	0.949	0.969	0.973	0.977	0.968	0.973	0.977	0.968
	M2	0.956	0.967	0.966	0.935	0.960	0.980	0.983	0.986	0.972	0.983	0.986	0.972
	M3	0.966	0.971	0.974	0.947	0.958	0.987	0.991	0.988	0.984	0.991	0.988	0.984
	M4	0.977	0.974	0.978	0.957	0.976	0.987	0.993	0.988	0.986	0.993	0.988	0.986
	M5	0.995	0.998	0.998	0.992	0.994	0.996	0.999	0.998	0.998	0.999	0.998	0.998

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

References

[1] P. Li, M. Yang, Q. Wu, et al., Confidence interval based distributionally robust real-time economic dispatch approach considering wind power accommodation risk, IEEE T Sustain. Energ. 12 (1) (Jan 2021) 58–69.
[2] G. Memarzadeh, F. Keynia, Short-term electricity load and price forecasting by a new optimal LSTM-NN based prediction algorithm, Electr. Pow. Syst. Res. 192 (Mar 2021) 106995.
[3] M. Jafari, Z. Malekjamshidi, J. Zhu, Design and development of a multi-winding high-frequency magnetic link for grid integration of residential renewable energy systems, Appl. Energy 242 (May 2019) 1209–1225.
[4] M. Cordeiro-Costas, D. Villanueva, P. Eguía-Oller, et al., Load forecasting with machine learning and deep learning methods, Appl. Sci. 13 (13) (2023) 7933.
[5] I. Ozer, S.B. Efe, H. Ozbay, A combined deep learning application for short-term load forecasting, Alexand. Eng. J. 60 (4) (2021) 3807–3818.
[6] J. Chen, L. Liu, K. Guo, S. Liu, D. He, Short-term electricity load forecasting based on improved data decomposition and hybrid deep-learning models, Appl. Sci. 14 (14) (2024) 5966.
[7] L. Weispfenning and A. Krontiris, "Review of Time Series Forecasting Methods and Error Evaluation: The Whole Process of Forecasting Exemplified With the German Electricity Consumption," 2024 18th International Conference on Probabilistic Methods Applied to Power Systems (PMAPS), Auckland, New Zealand, 2024, pp. 1-6.
[8] C.R. Deng, X.Y. Zhang, Y.M. Huang, et al., Equipping seasonal exponential smoothing models with particle swarm optimization algorithm for electricity consumption forecasting, Energies. (Basel) 14 (13) (Jul 2021) 4036.
[9] P. Almaleck, S. Massucco, G. Mosaico, et al., Electrical consumption forecasting in sports venues: A proposed approach based on neural networks and ARIMAX models, Sustain. Cities. Soc. 100 (2024) 105019.
[10] I. Kochetkova, et al., Short-term mobile network traffic forecasting using seasonal ARIMA and Holt–Winters models, Future Int. 15 (9) (2023) 290.

[11] Q.B. Ge, C. Guo, H.Y. Jiang, et al., Industrial Power Load Forecasting Method Based on Reinforcement Learning and PSO-LSSVM, IEEE T Cybernet. 52 (2) (Feb 2022) 1112–1124.
[12] T.Y. Li, Z.J. Qian, T. He, Short-Term Load Forecasting with Improved CEEMDAN and GWO-Based Multiple Kernel ELM, Complexity. 2020 (Feb 2020) 1209547.
[13] G.F. Fan, L.Z. Zhang, M. Yu, et al., Applications of random forest in multivariable response surface for short-term load forecasting, Int. J. Elec. Power 139 (Jul 2022) 108073.
[14] C. Zhang, D. Hu, and T. Yang, "Anomaly detection and diagnosis for wind turbines using long short-term memory-based stacked denoising autoencoders and XGBoost," Reliab. Eng. Syst. Safe., vol. 222, pp. 108445, Jun 2022.
[15] E. Mocanu, D.C. Mocanu, P.H. Nguyen, A. Liotta, M.E. Webber, M. Gibescu, J. G. Slootweg, On-line building energy optimization using deep reinforcement learning, IEEe Trans. Smart. Grid. 10 (4) (Jul. 2019) 3698–3708.
[16] Z.Y. Yin, X.Q. Ji, Y.M. Zhang, et al., Data-driven approach for real-time distribution network reconfiguration, IET Gener Transm Dis 14 (13) (Jul 2020) 2450–2463.
[17] X.Q. Ji, Z.Y. Yin, Y.M. Zhang, et al., Real-time robust forecasting-aided state estimation of power system based on data-driven models, Int. J. Elec. Power 125 (Feb 2021) 106412.
[18] M. Tan, S.P. Yuan, S.H. Li, et al., Ultra-Short-Term Industrial Power Demand Forecasting Using LSTM Based Hybrid Ensemble Learning, IEEE T Power Syst. 35 (4) (Jul 2020) 2937–2948.
[19] Y. Lin, M. Yang, C. Wan, et al., A multi-model combination approach for probabilistic wind power forecasting, IEEE T Sustain. Energ. 10 (01) (Jan 2019) 226–237.
[20] H.H. Bian, Y.Q. Zhong, J.S. Sun, et al., Study on power consumption load forecast based on K-means clustering and FCM-BP model, Energy Reports 6 (Dec 2020) 693–700.
[21] S. Smyl, G. Dudek, P. Pełka, ES-dRNN: a hybrid exponential smoothing and dilated recurrent neural network model for short-term load forecasting, IEEe Trans. Neural Netw. Learn. Syst. 35 (8) (Aug. 2024) 11346–11358.
[22] R. Bareth, A. Yadav, S. Gupta, M. Pazoki, Daily average load demand forecasting using LSTM model based on historical load trends, IET Gener. Transm. Distrib 18 (5) (2024) 952–962.
[23] M. Alhussein, K. Aurangzeb, S.I. Haider, Hybrid CNN–LSTM model for short-term individual household load forecasting, IEEe Access. 8 (2020) 180544–180557.
[24] G.J. Wang, X.H. Wang, Z.X. Wang, et al., A VMD-CISSA-LSSVM based electricity load forecasting model, Mathematics 10 (1) (Jan 2022) 28.
[25] Y. Wang, H. Li, A. Jahanger, Q. Li, B. Wang, D. Balsalobre-Lorente, A novel ensemble electricity load forecasting system based on a decomposition-selection-optimization strategy, Energy 312 (2024) 133524.
[26] G.F. Fan, Y. Li, X.Y. Zhang, et al., Short-term load forecasting based on a generalized regression neural network optimized by an improved sparrow search algorithm using the empirical wavelet decomposition method, Energy Sci. Eng. (Apr 2023) (Early Access).
[27] H. Dehghani, B. Vahidi, S.H. Hosseinian, Wind farms participation in electricity markets considering uncertainties, Renew. Energy 101 (Feb. 2017) 907–918.
[28] C.D. Fan, C.K. Ding, J.H. Zheng, et al., Empirical Mode Decomposition based Multi-objective Deep Belief Network for short-term power load forecasting, Neurocomputing. 388 (May 2020) 110–123.
[29] G.F. Fan, Y.R. Liu, H.Z. Wei, et al., The new hybrid approaches to forecasting short-term electricity load, Electr. Pow. Syst. Res. 213 (Dec 2022) 108759.
[30] M.A. Colominas, G. Schlotthauer, M.E. Torres, Improved complete ensemble EMD: A suitable tool for biomedical signal processing, Biomed. Signal. Process. Control 14 (2014) 19–29.
[31] L. Dong, N. Xu, P. Zhang, et al., An ICEEMDAN–WPD based denoising method for MS signals and its engineering application, Nondestruct. Test. Eval. (2024) 1–23.
[32] W. Li, Q. Shi, M. Sibtain, et al., A hybrid forecasting model for short-term power load based on sample entropy, two-phase decomposition and whale algorithm optimized support vector regression, IEEe Access. 8 (2020) 166907–166921.
[33] S. Huang, J. Zhang, Y. He, et al., Short-term load forecasting based on the CEEMDAN–sample entropy–BPNN–transformer, Energies. (Basel) 15 (10) (2022) 3659.
[34] A. Wan, Q. Chang, A.L.B. Khalil, et al., Short-term power load forecasting for combined heat and power using CNN–LSTM enhanced by attention mechanism, Energy 282 (2023) 128274.
[35] A. Sherstinsky, Fundamentals of recurrent neural network (RNN) and long short-term memory (LSTM) network, Phys. D: Nonl. Phenomena 404 (2020) 132306.
[36] F. M. Salem and F. M. Salem, "Recurrent neural networks (RNN)," in Recurrent Neural Networks: From Simple to Gated Architectures, 2022, pp. 43–67.
[37] B. Lindemann, T. Müller, H. Vietz, et al., A survey on long short-term memory networks for time series prediction, Procedia CIRP. 99 (2021) 650–655.
[38] N. Son, S. Yang, J. Na, Deep neural network and long short-term memory for electric power load forecasting, Appl. Sci. 10 (18) (2020) 6489.